

Viva Voce Examination Questions

Introduction

Q: Why did you choose to explore LUPI instead of another technique for improving object detection?

A: We chose to investigate the LUPI paradigm because its application in object detection remains underexplored, making it both a novel and timely research direction. We hypothesised that incorporating privileged information during training could enhance model performance while still keeping the models lightweight and practical for deployment. Unlike other approaches, LUPI leverages information only available at training time, which improves the learning process without increasing inference complexity or requiring additional annotations. This gave us a controlled and generalisable way to test improvements while staying aligned with the goal of efficiency and real-world applicability. **[Remember Konstantinos' point on domain-specific models, which require large datasets, and thus, that is also why we applied for LUPI. LUPI is to leverage highly informative streams when we don't have large datasets, and therefore, use smaller datasets.]**

Q: Why did you choose UAV-based litter detection as your main case study?

A: The UAV-based litter detection case study was chosen because it represents a particularly challenging object detection task, as the litter items are small, diverse in appearance, and often occur in cluttered real-world environments. To ensure a comprehensive evaluation, we included all available datasets for this task, which allowed us to demonstrate the robustness of the proposed approach. Importantly, the case study also provided a practical application context, where lightweight models are essential for deployment on UAVs. This aligns with the motivation of our work, as the method can be integrated into real-time UAV systems and has potential extensions to traffic monitoring and similar lightweight, resource-constrained environments.

Q: Your dissertation cites the publication "MRTMD: A Multi-Resolution Dataset for Evaluating Object Detection in Traffic Monitoring Systems" as a publication of the dissertation. How does this paper relate to your dissertation?

A: That publication laid important groundwork for this dissertation in two main ways. First, it explored the deployment of object detection systems in a practical and lightweight context, specifically for traffic monitoring. This aligns with the overarching aim of my dissertation, which is to develop techniques that can be efficiently deployed in real-world scenarios. Second, it involved the use of renowned object detection models, giving me direct experience with their behaviour, limitations, and potential when applied in challenging environments. While that work did not involve LUPI directly, it provided a solid foundation in terms of dataset design, evaluation methodologies, and the practical aspects of object detection, which informed and supported the direction of this dissertation.

Background and Literature Review

Q: *There are at least two prior works that explored LUPI for object localisation (Feyereisl et al., 2014; Sun et al., 2018). Why did you not adopt their approaches or extend directly from them?*

A: Both of those studies are valuable early explorations of the LUPI paradigm in computer vision, but there are several reasons why their approaches were not directly adopted in this dissertation. First, they are relatively outdated, relying on algorithms such as SVMs and structural SVM+, which do not align with current SOTA deep learning-based object detection pipelines. Since then, significantly more powerful object detection architectures have been developed, which motivated us to explore LUPI in this modern context. Second, both papers use segmentation masks as privileged information. As explained in the methodology, segmentation masks are annotation-intensive and less widely available compared to bounding box annotations. They also introduce practical challenges such as overlap handling and additional error surfaces when predicted from bounding boxes. Third, their scope was limited to object localisation on the Caltech-UCSD Birds 2011 dataset, with single-class objects in relatively constrained contexts. In contrast, our work investigates multi-class object detection in more complex and challenging real-world scenarios. For these reasons, rather than replicating older approaches, this study builds on the idea of LUPI and adapts it to contemporary object detection tasks and datasets, making the contribution more relevant and impactful.

Q: *Why does a substantial portion of your background focus on reviewing object detection techniques and litter detection approaches, and datasets, rather than approaches to LUPI in computer vision?*

A: The balance of the background chapter was deliberate (intentional). Since the dissertation applies LUPI within the context of object detection and UAV-based litter detection, it was necessary to review these areas in depth to set the stage for the practical challenges and experimental design discussed in later chapters. Object detection has a rich literature with numerous architectures and benchmarks, and litter detection is a relatively new but practically important domain with its own datasets and constraints, so reviewing these fields provided the foundation for understanding the scope and motivation of this work. By contrast, the literature on LUPI in computer vision is relatively limited, especially in object detection, so while it was reviewed, it could not form as substantial a section. In short, the review structure reflects both the practical focus of the dissertation and the relative availability of prior work in each area.

Methodology

Q: Why didn't you use polygon or segmentation masks for privileged information, as seen in previous literature?

A: While polygon or segmentation masks are often employed in the literature because they provide more precise object representations, we deliberately chose bounding box masks for two key reasons. First, most object detection datasets are annotated with bounding boxes rather than polygons, meaning polygon masks would need to be generated using a segmentation model. These predictions cannot be considered ground truth and would introduce an additional error surface. Second, bounding boxes offer a more controlled error margin and reduce unnecessary complexity, especially when dealing with overlapping objects, where polygons make it more difficult to consistently identify object centres (easily defined object corners). Rather than fully relying on the literature, we questioned whether polygon masks were truly necessary in this context and found that bounding boxes offered a more practical and robust choice. Some of our very early preliminary tests with polygon masks highlighted these issues. Nonetheless, we recognise that polygon or segmentation masks remain a promising avenue for future work, particularly in domains where such annotations are more readily available.

Q: Why did you use single-channel grayscale images as privileged information instead of RGB or other representations?

A: We chose single-channel grayscale images primarily for compactness and computational efficiency, which is especially important for lightweight and practical deployment scenarios. Compared to RGB or other multi-channel representations, grayscale reduces memory and processing requirements while still capturing essential features. While segmentation masks have been explored in prior work, it was only once and specifically for object localisation, not full object detection. Using grayscale images represents a novel proof of concept, allowing us to focus on demonstrating the core hypothesis. Although alternative representations might offer different advantages, this approach successfully validated that incorporating privileged information can improve model performance in a controlled and efficient manner.

Q: Did you consider other forms of privileged information besides bounding box masks?

A: Yes, I experimented with depth maps, saliency maps, and even combinations of the two (You can also mention HoG, Texture Maps, Corner and Edge Detection). These alternatives provided only marginal improvements. By contrast, the bounding box mask as privileged input gave much more significant improvements. This is because the mask directly encodes the spatial location of the objects, giving the teacher model a strong signal about where the objects of interest lie. That spatial guidance was crucial in enabling the teacher to achieve high accuracy, which in turn made the knowledge distillation process effective. Without the teacher acquiring sufficiently precise representations, the student would not have been able to benefit to the same degree. Refer to Appendix A.

Q: Why didn't you implement YOLO or transformer-based detectors, and why did you only choose Faster R-CNN as a two-stage detector?

A: The focus of this dissertation was to evaluate the proposed LUPi approach across a representative set of object detection architectures while keeping the experimental setup manageable and consistent. We chose five well-established models – Faster R-CNN, SSD, SSDLite, RetinaNet, and FCOS – which cover both two-stage and single-stage paradigms and provide a strong benchmark for comparison. YOLO and transformer-based detectors, while popular and powerful, were omitted because implementing them would have added significant complexity and time overhead, potentially affecting other critical components of the dissertation, such as dataset preprocessing, training pipelines, and evaluation consistency. Faster R-CNN was selected as the sole two-stage detector because it is the canonical, widely recognised two-stage architecture that represents region proposal-based detection. Including additional two-stage detectors would have increased experimental complexity without substantially altering the conclusions. We acknowledge this as a limitation, and future work could extend the evaluation to include YOLO, transformer-based models, or other two-stage architectures to further assess the generalisability of the proposed approach.

Q: Why did you choose the final backbone layer for feature-level distillation?

A: For feature-level distillation, it was important to select a layer that captures semantically rich features and can be directly compared between the teacher and student networks. We chose the final backbone layer because it is widely used in existing studies and provides high-level representations that are well-suited for knowledge transfer. Specifically, for Faster R-CNN, RetinaNet, and FCOS with ResNet-FPN backbones, this corresponds to the last convolutional layer before the FPN, while for SSD and SSDLite, it is the final layer before the auxiliary layers and detection heads in VGG-16 and MobileNetv3. While other layers or combinations of layers could potentially improve distillation, we focused on the final backbone layer for consistency and practical reasons across all experiments. Investigating alternative or intermediate layers remains an interesting avenue for future work.

Q: Why didn't you freeze parts of the model during training and instead fine-tune the entire network?

A: In this study, we fine-tuned the entire model rather than freezing layers, which is sometimes done in object detection training. Preliminary experiments with freezing parts of the model showed worse performance, likely because the network could not fully adapt to the privileged information. Since we had sufficient computational resources, we opted for full fine-tuning to allow both the backbone and detection heads to optimally learn from the LUPi approach and the specific datasets used.

Q: Did you explore distillation between different architectures, such as SSD teacher and Faster R-CNN student? And did you try distillation at multiple layers?

A: Distillation between different architectures, such as SSD as a teacher and Faster R-CNN as a student, was considered but is out of the scope of this study. Since this is the first major study applying

LUPI to object detection with modern deep learning architectures, we focused on a controlled setup where the teacher and student share the same architecture. Similarly, while distillation at multiple layers could potentially improve knowledge transfer, we concentrated on the final backbone layer for consistency across all models and practical feasibility. This approach ensures that any observed improvements can be confidently attributed to the use of privileged information rather than architectural differences or experimental complexity. Exploring cross-architecture distillation and multi-layer distillation remains a promising direction for future work.

Q: Why did you use cosine distance for feature-level distillation instead of, for example, Euclidean distance?

A: Cosine distance, as used in the Lab2Wild (Konstantinos et al.) paper, computes the similarity or dissimilarity between two vectors of latent feature representations and is particularly suitable when comparing unit-normalised vectors. While Euclidean distance is an alternative, cosine distance is dominant in the literature for feature-level distillation because it focuses on the orientation of the vectors rather than their magnitude, which helps capture semantic similarity more effectively. This makes it particularly well-suited for knowledge transfer between teacher and student networks.

Q: How did you choose the alpha values for your experiments?

A: The alpha values were chosen to uniformly cover the full range from 0 to 1, following the approach in Lab2Wild (Konstantinos et al.) and other literature. Specifically, we used [0, 0.25, 0.5, 0.75, 1] to ensure a consistent and systematic evaluation across the spectrum, allowing us to observe the effect of different weightings between the teacher and student contributions in the distillation process. 0 denotes baseline.

Q: Why did you use relatively simple training parameters and preprocessing steps instead of performing extensive hyperparameter tuning?

A: The focus of this study was to evaluate the effectiveness of the proposed approach across multiple object detection architectures in a fair and controlled way. To ensure consistency between the baselines and our method, we used standard training parameters and preprocessing steps without extensive hyperparameter tuning. The aim was not to maximise absolute accuracy for each model but to demonstrate the improvements introduced by the approach in a reproducible setup. Hyperparameter optimisation could certainly enhance performance and is a potential direction for future work, but it was outside the scope of this dissertation.

Q: Why did you choose to resize images to 800 pixels, and why did you use min-max normalisation?

A: We resized images to 800 pixels as a compromise across all detectors. This value represents an average input size that works well with the smallest input required by the ResNet-FPN backbone, while SSD and SSDLite can handle internal resizing to 300 and 332, respectively. Min-max normalisation was applied because the torchvision models expect input images in the $[0-1]$ range, which is standard practice in computer vision models and ensures consistent scaling across all experiments.

Q: Why was the UAVVaste dataset not tiled, while models trained on SODA were tiled?

A: The UAVVaste dataset was not tiled because the literature and standard preprocessing for this dataset do not specify tiling. In contrast, SODA images contain objects that appear at larger and more varied heights, so tiling was necessary to ensure that these objects could be effectively processed by the network and to achieve a fair evaluation. Following the established preprocessing conventions for each dataset ensured consistency with prior work and a controlled experimental setup.

Q: Why did you focus on experiments with binary litter detection?

A: We focused on binary litter detection because most prior studies in the literature use this setup, which provides a clear baseline for comparison. Conducting experiments with binary classes allowed us to evaluate the core effectiveness of the proposed LUPI approach in a controlled and interpretable manner before extending to multi-class scenarios. Additionally, it enabled us to assess generalisability on how increasing class granularity impacts model performance, providing insights into both binary and multi-class setups.

Evaluation

Q: How does your proposed approach generalise across datasets and classes?

A: We have shown that the approach generalises well to up to 20 classes and performs robustly across different datasets. Both within-dataset and across-dataset experiments were conducted, using challenging scenarios and popular object detection metrics to validate performance. However, factors that hinder generalisation include (i) overlapping objects of the same type, (ii) occlusion of smaller objects by larger ones, and (iii) insufficient colour representations in datasets with many classes. Despite these challenges, the method consistently demonstrates improved detection accuracy and robustness compared to baseline models.

Q: Why do some people consider your results “poor” compared to state-of-the-art models, and why should your work be considered if it doesn't improve SOTA performance?

A: The models we used were popular and well-established, but not necessarily the latest state-of-the-art. The goal of this study was not to surpass SOTA performance but to demonstrate the effectiveness of the LUPI paradigm in object detection across multiple architectures. By keeping the baselines consistent and controlled, we could isolate the impact of privileged information, showing meaningful improvements relative to the baseline, which validates the proposed methodology even if absolute SOTA performance was not achieved.

Q: You claim that LUPI improves accuracy without increasing model size, parameters, or inference time. How do you support this claim?

A: We validated this by comparing both the baseline and student models with and without LUPI. The comparisons show that model size, number of parameters, and GFLOPs remain unchanged, while inference speed (FPS) is maintained. At the same time, accuracy improves with the addition of privileged information, demonstrating that LUPI provides performance boosts without any computational or memory overhead at runtime.

Q: Why did Faster R-CNN, RetinaNet, and FCOS perform better on the UAV-based litter detection datasets but worse on Pascal VOC 2012, while SSD and SSDLite showed the opposite trend?

A: The difference in performance can be attributed to the nature of the datasets and the model architectures. Litter datasets often contain smaller objects at varied heights and resolutions, where ResNet-FPN-based architectures like Faster R-CNN, RetinaNet, and FCOS excel. In contrast, Pascal VOC images are generally smaller and contain more overlapping objects, which are largely defined. Since Faster R-CNN and similar models rely on region proposal networks (RPNs), the privileged information can sometimes confuse proposal generation in dense or overlapping scenes, reducing accuracy. Additionally, these models were initialised with pre-trained COCO weights, which assume

higher-resolution images, further contributing to poorer performance on Pascal VOC. On the other hand, SSD and SSDLite are designed for lower-resolution inputs and are more robust in these scenarios, explaining their superior performance. Finally, all models were run with the same configurations for fairness, so differences in initialisation, image resolution, and training parameters also influenced the observed results.

Q: How can you claim that your proposed approach improves small object detection?

A: The improvement in small object detection is demonstrated through aggregated metrics such as mAP and mAR across all test datasets. In addition, we conducted an object-size-based performance analysis using COCO-style metrics, including mAP and mAR for small, medium, and large objects. This detailed evaluation shows that the proposed approach particularly enhances performance on small objects. Segregated results by object size consistently indicate better detection accuracy for small objects compared to the baseline models, confirming the effectiveness of the method in handling challenging small-object scenarios.

Q: Why didn't you evaluate your approach on the COCO dataset?

A: Our primary focus was on evaluating the LUPI-based approach in practical, challenging scenarios, such as UAV-based litter detection, which involves small objects and real-world variability. To demonstrate generalisability beyond a single domain, we also included an experiment on the Pascal VOC 2012 dataset. Whilst the COCO dataset could have been explored, the goal was not to maximise performance on every large-scale benchmark but rather to validate the approach across multiple architectures and domains in a manageable and reproducible setup. Extending the evaluation to COCO and other large-scale datasets remains a logical avenue for future work.

Q: Why did you only present interpretability results for the SODA 1-metre dataset?

A: Interpretability tests were carried out across all selected architectures and datasets, focusing on the specific target layers where distillation took place to examine differences in feature-based learning between baseline and student models. We presented only the results from the SODA 1-metre dataset for brevity because it showed the most significant improvements under the proposed LUPI-based approach. The other datasets yielded relatively similar patterns, reinforcing the conclusion that LUPI helps the model shift focus from irrelevant background features to objects of interest (or else dissipating attention across the entire image in complex scenes), which is particularly important in challenging backgrounds. Presenting this subset allows for a clear illustration of the effect without redundancy, while still supporting the overall findings.

Q: What are the limitations of using privileged information in your study?

A: The limitations of using privileged information can be grouped into three main challenges: (i) overlapping objects of the same type, (ii) large objects obscuring smaller ones, and (iii) difficulties distinguishing between object classes in datasets with many categories. Additionally, training with privileged information requires more time compared to standard training without it. However, since inference is typically performed far more frequently than training in practical deployment scenarios, this trade-off is considered both practical and worthwhile. Despite these limitations, privileged information still provides a meaningful improvement in guiding the model to focus on relevant features.

Q: Why did you choose a 3×3 tiling instead of a 5×5 (done in the original SODA paper) or explore other tiling options?

A: While previous literature often applied tiling, the authors did not justify their choice. In our study, a 3×3 tiling was selected based on experimental results as a practical balance between accuracy and efficiency. Although 5×5 tiling could improve accuracy by reducing small-object issues, it would drastically increase the number of images to process. The 3×3 tiling choice strikes a compromise, enhancing performance without overly increasing computational load.

Q: Why didn't you compare your results directly with other literature for each dataset and model?

A: The focus of this study was on evaluating the improvements introduced by the proposed LUPI approach across multiple renowned object detection architectures, rather than matching or surpassing every published result. We used standard implementations of the same model types as baselines to provide a fair and controlled benchmarking environment. While comparing with literature could be interesting, it was outside the scope of this dissertation, since the primary goal was to demonstrate the effectiveness and generality of the approach in a reproducible and well-documented setup.

Conclusion

Q: What is the main conclusion of your dissertation regarding LUPI in object detection?

A: The research question was whether LUPI can be effectively integrated into object detection models and how feasible and generalisable such an approach is. This dissertation is not application-oriented and is not aimed at improving litter detection specifically. Instead, it proposes a new methodology applicable both within litter detection and more broadly in object detection. To validate this, we used baseline models across multiple architectures and datasets, observing consistent improvements. We can therefore conclude that the proposed methodology successfully enhances model performance while being feasible and generalisable.

Q: In what applications could your proposed LUPI-based methodology be used?

A: The methodology improves detection accuracy without increasing model size or inference time, making it broadly applicable to scenarios where lightweight, efficient models are needed. This includes (1) UAV-based litter detection, where small objects must be located quickly across complex terrains (such as the AWIGS Project); (2) traffic monitoring systems, which require real-time detection under varying conditions (connect to Mark's paper); (3) video surveillance on resource-constrained devices; and (4) underwater exploration, where limited visibility and onboard processing demand compact yet accurate models.

Q: What are the main limitations of your LUPI-based approach?

A: In this dissertation, several limitations were observed. Within the privileged information generation, challenges such as (1) overlapping objects of the same type, (2) occlusion of smaller objects by larger ones, and (3) difficulties distinguishing classes in datasets with many categories affected performance. More generally, (4) the teacher-student setup and use of privileged information increases training time, which may be a concern for resource-constrained or time-sensitive deployments. (5) Privileged data is not always readily available and may require labour-intensive construction or simulation, and its effectiveness depends on careful task-specific design since no standard formats exist for many datasets. (6) While LUPI improves accuracy without increasing model size, parameters, or inference time, these benefits must be balanced against the additional complexity during training and the effort needed to generate and validate privileged information. Overall, these limitations highlight trade-offs between performance gains and practical applicability.

Q: What are the potential directions for future work based on your dissertation?

A: Several directions could extend this work. (1) The methodology could be applied to more recent state-of-the-art object detection frameworks such as YOLOv12, DETR, or RT-DETR to test generalisability and uncover new strengths or limitations. (2) Privileged information could be

represented in more advanced ways, for example, using semantic maps, attention hints, or other structured cues to better capture object relationships and context. (3) Knowledge distillation could be refined to influence both classification and regression branches, which may improve the detection of harder-to-detect objects. (4) Architecture-specific optimisation, including hyperparameter tuning, data augmentation, and loss function adjustments, could further enhance performance. (5) Finally, the method could be extended to dense prediction tasks like semantic or instance segmentation, where privileged information may help the network learn subtle spatial cues without increasing inference complexity.

General Questions

Q: *If you could have done something differently in this study, what would it be?*

A: *If time allowed, I would have explored more variations of privileged information and additional knowledge distillation methods and deep learning architectures to see if further improvements could be achieved. However, doing so would have reduced the time available for other important components, such as implementing and evaluating five different object detection models, as well as conducting a large number of experiments. Expanding the study in this way could make it broader in terms of exploration, but might have weakened the generalisability and the strength of the claims demonstrated in this dissertation.*
